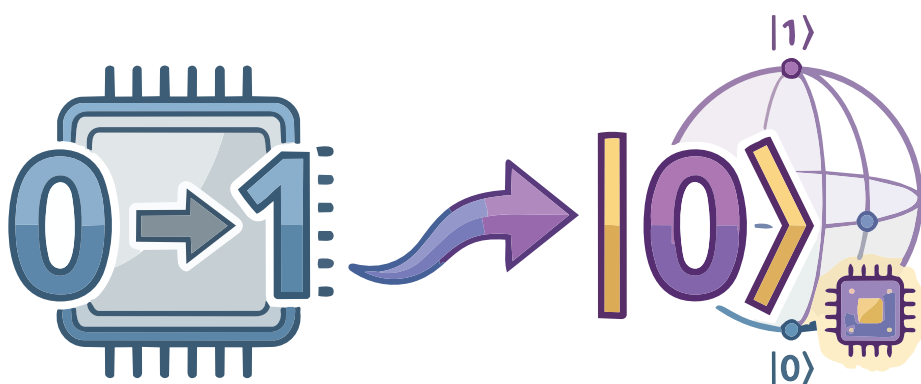


Quantum Computing:

From Zero to Ket Zero

This series on quantum computing aims to equip beginners with a strong conceptual foundation and gradually guide them towards an intermediate-level understanding of quantum computing, enabling them to participate confidently in this emerging tech revolution.



CLASSICAL BIT

QUANTUM QUBIT

The year 2025 was observed by the United Nations as the International Year of Quantum Science and Technology (IYQ) to commemorate one hundred years of quantum mechanics. The quantum era began with the publication of the celebrated research paper known as the ‘Umdeutung’ paper by the renowned physicist and Nobel laureate Werner Heisenberg. Over the past century, quantum mechanics has influenced nearly every aspect of modern life by providing the scientific foundation for technologies such as semiconductors, integrated circuits, MRI scanners, lasers, electron microscopes, and GPS systems. Yet, it is quite possible that the greatest technological achievement enabled by quantum mechanics is still to come: the quantum computer.

Interest in quantum computing has grown dramatically in recent years, supported by substantial investments

from both technology companies and governments around the world. The emergence of quantum machine learning has further strengthened the case for developing scalable and reliable quantum hardware. Major technology companies—including IBM, Google, Microsoft, NVIDIA, and Meta—are investing heavily in quantum technologies and related research.

The government of India launched the National Quantum Mission (NQM) in April 2023 under the Department of Science and Technology to accelerate the development of quantum technologies in the country. Many other nations are also investing heavily in quantum computing. This global surge of interest is hardly surprising. It is widely believed that the first nation to develop a large-scale, fault-tolerant quantum computer could gain a substantial strategic and technological advantage. This is similar to the military

advantage of a country that alone possesses nuclear weapons. It is easy to see that such a nation would wield enormous influence and effectively dictate the rules of the world.

This brings us to an important question for students of science and technology, as well as IT professionals: should we strive to become early adopters who help shape the future of quantum computing, or should we wait and follow developments led by others?

Many existing tutorials treat quantum programming as just another programming subject, focusing mainly on writing code while paying little attention to the mathematics behind the code or the hardware that ultimately executes it. The distinctive feature of this series is its three-level approach to understanding quantum computing:

- The coding level — what happens when a particular line of quantum code is written.